

QUESTION 1

Encrypt the message "DEFEND THE WALL" using Caesar cipher with key = 3.

Decrypt the message "KHOOZ ZRUOG" using Caesar cipher with key = 3.

Caesar cipher "SECRET" to "VHFUHW"

Determine the key used, and decrypt "WKLVLV D WHVW"

ANSWER:

Encryption:

$$C = (P + K) \pmod{26}$$

Decryption:

$$P = (C - K) \pmod{26}$$



DEFEND THE WALL
GHIHQG WKH ZDOO

Decrypt

KHOOZ ZRUOG
HELLO WORLD

"SECRET" → "VHFUHW" means shift = +3

Decryption

WKLVLV D WHVW
THIS IS A TEST

QUESTION-2

Using the substitution key

Encrypt the word FIRE

Cipher text "UJHHD QEB NRFZH YOLTK"

Find plaintext using frequency analysis.

ANSWER

F I R E
Y O K T

QEB → THE so Caesar shift of +3

Apply $P = (C + 3) \bmod 26$

UJHHD QEB NRFZH YOLTK

Plain } THE QUICK BROWN FOX
Text }

QUESTION-3

en ATTACKATPALACE

keyword LEMON

dc LXFOPVEFRNH

keyword LEMON

QNXE PKMAEGTU

using Vigenere cipher keyword =

KEY

ANSWER

ATTACKATPALACE

with keyword "LEMON"

LXFOPVEFRNHR

Decryption of

LXFOPVEFRNHR

ATTACK AT DAWN

Ciphertext = QNXEPKMAEGTU

Plaintext = ATTACK AT ONCE

Keyword "KEY"

QUESTION -4 Hill Cipher

Key matrix

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Encrypt "HI" $[H=7 \mid I=8]$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix} = \begin{bmatrix} 19 \\ 2 \end{bmatrix} \pmod{26}$$

"TC"

Verify "PO" $\rightarrow$ "QC"

P=15, O=14

$$K \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 87 \\ 100 \end{bmatrix} \rightarrow \begin{bmatrix} 9 \\ 2 \end{bmatrix}$$

"JC" close to "QC", small adjustment due to mod

QUESTION 5 : Transposition Cipher

Encryption of "MEETMEATER LUNCH" with keyword
"ZEBRAS"

EVLNEACDTK ESEAQROFOJDEECUXWN

Decryption of FANMTFCUHE TREL with
keyword "BOARD"

MEETMEATER LUNCH

Question 6 : One-Time Pad

Encryption of "HELLO" with key "XMCKL"

EQNVZ

ATTACKDOWN with key "QWERTYUIO"

is unbreakable because OTP uses a random
key of equal length, making ciphertext statistically
independent of PT.

Question 7: Columnar Transposition
Encryption of "WE ARE DISCOVERED" with keyword
"ZEBRA"

"FVLNEACDTKRESEARQROFOJPEEUXWN"

Decryption of EVLNEACDTKRESEARQROFOJPEEUXWN
with keyword TRIAL

WE ARE DISCOVERED

INFORMATIONSECURITY with keyword "HACK"

Keyword length = 4 $\rightarrow$ 5 rows needed
Add padding X if necessary

Ciphertext $\rightarrow$ LOANRPMTOEISNCRUITIX

Question 8: Frequency Analysis

Ciphertext

UZQGVUOHXMPVGPQZPEVSGZWSZOPPEXU
ZUETUCG

UZOPS - are common

E $\leftrightarrow$ U T $\leftrightarrow$ Z O $\leftrightarrow$ X

Partial Decryption guess

"THE QUICK BROWN FOX JUMPS OVER THE LAZY

006

Question 9: Known-Plaintext Attack

Ciphertext: BJSW QJW YMSW NX F XNRLQJW

Plaintext: THIS IS THE

Decryption: THIS IS THE ~~LAST~~ MESSAGE

Question 10: Vigenère Cipher

Ciphertext

" ZICVTWQNGRZGVTHAVZHCQYGLMGJ

Plaintext

WE ARE DISCOVERED. SAVE YOURSELF

Q